MakeItFrom.com Menu (ESC)

Home > Iron Alloy > Wrought Precipitation-Hardening Stainless Steel > S45500 Stainless Steel

Annealed S45500 Stainless Steel

Annealed S45500 stainless steel is S45500 stainless steel in the annealed condition. It has the lowest strength and lowest ductility compared to the other variants of S45500 stainless steel.

The graph bars on the material properties cards below compare annealed S45500 stainless steel to: wrought precipitation-hardening stainless steels (top), all iron alloys (middle), and the entire database (bottom). A full bar means this is the highest value in the relevant set. A half-full bar means it's 50% of the highest, and so on.

Mechanical Properties

Brinell Hardness

280

Elastic (Young's, Tensile) Modulus

190 GPa 28 x 10^6 psi

Elongation at Break

3.4 %

Fatigue Strength

570 MPa | 83 x 10^3 psi

Poisson's Ratio

0.28

Rockwell C Hardness

32

Shear Modulus

75 GPa | 11 x 10^6 psi

Shear Strength

790 MPa | 110 x 10^3 psi

Tensile Strength: Ultimate (UTS)

1370 MPa | 200 x 10^3 psi

Tensile Strength: Yield (Proof)

1240 MPa | 180 x 10^3 psi

Thermal Properties

Latent Heat of Fusion

270 J/g

Maximum Temperature: Corrosion

620 °C 1140 °F

Maximum Temperature: Mechanical

760 °C 1400 °F

Melting Completion (Liquidus)

1440 °C 2630 °F

Melting Onset (Solidus)

1400 °C 2550 °F

Specific Heat Capacity

470 J/kg-K 0.11 BTU/lb-°F

Thermal Expansion

11 µm/m-K

Otherwise Unclassified Properties

Base Metal Price

17 % relative

Density

7.9 g/cm^3 490 lb/ft^3

Embodied Carbon

3.8 kg CO_2/kg material

Embodied Energy

57 MJ/kg 24 x 10^3 BTU/lb

Embodied Water

120 L/kg 14 gal/lb

Common Calculations

PREN (Pitting Resistance)

13

Resilience: Ultimate (Unit Rupture Work)

45 MJ/m^3

Resilience: Unit (Modulus of Resilience)

4000 kJ/m^3

Stiffness to Weight: Axial

14 points

Stiffness to Weight: Bending

24 points

Strength to Weight: Axial

48 points

Strength to Weight: Bending

35 points

Thermal Shock Resistance

48 points

Alloy Composition

Iron (Fe)	71.5 to 79.2
Chromium (Cr)	11 to 12.5
Nickel (Ni)	7.5 to 9.5
Copper (Cu)	1.5 to 2.5
Titanium (Ti)	0.8 to 1.4
Manganese (Mn)	0 to 0.5
Silicon (Si)	0 to 0.5
Molybdenum (Mo)	0 to 0.5
Niobium (Nb)	0 to 0.5
Tantalum (Ta)	0 to 0.5
Carbon (C)	0 to 0.050
Phosphorus (P)	0 to 0.040
Sulfur (S)	0 to 0.030

All values are % weight. Ranges represent what is permitted under applicable standards.

Followup Questions

How are the material properties defined?

How do I compare this exact variant to another material?

Further Reading

ASTM A693: Standard Specification for Precipitation-Hardening Stainless and Heat-Resisting Steel Plate, Sheet, and Strip

Welding Metallurgy of Stainless Steels, Erich Folkhard et al., 2012

ASTM A959: Standard Guide for Specifying Harmonized Standard Grade Compositions for Wrought Stainless Steels

Properties and Selection: Irons, Steels and High Performance Alloys, ASM Handbook vol. 1, ASM International, 1993

Corrosion of Stainless Steels, A. John Sedriks, 1996

ASM Specialty Handbook: Stainless Steels, J. R. Davis (editor), 1994

Advances in Stainless Steels, Baldev Raj et al. (editors), 2010